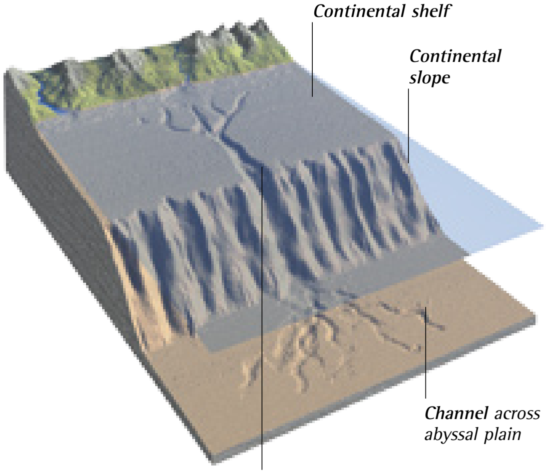
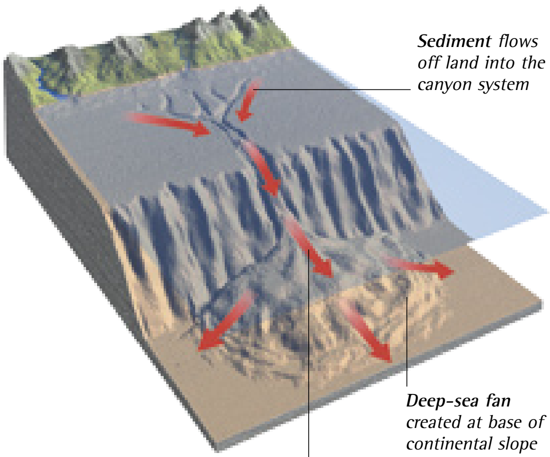
ABYSSAL PLAINS

The rocks of continents are constantly worn away by wind, frost, and rain, as well as by coastal erosion. Gravel, sand, and mud are carried off the land by rivers and swept into the sea. These sediments fan out from river mouths in the form of deep deposits on the seabed. Known as submarine fans, some cover huge areas and distort the Earth's crust with their weight. The sediments flow down canyons in the continental slope at the edge of the continental shelf, forming deep deposits at the foot of the slope. They spread out to form abyssal plains, which are vast flat areas of soft sediments on the ocean floors. These sediments also contain windblown dust particles, mud, and sand dropped by melting icebergs, and the skeletons of countless microscopic marine organisms that sink to the ocean floor to form thick layers of soft ooze. Over time the soft sediments harden into sedimentary rocks, and millions of years later major earth movements may raise these rocks to form new land.



SUBMARINE CANYON IN CONTINENTAL SHELF



TURBIDITY CURRENT AND DEEP-SEA FAN

▲ MUDFLOWS AND CANYONS
Rock, sand, silt, and mud can cascade off the continental shelf in underwater avalanches called turbidity currents, especially during storms and earthquakes. These dense, heavy flows can scour deep canyons in the fringes of the continental shelf. Off the eastern coast of the US, for example, there are canyons 2,625 ft (800 m) deep. Submarine fans of debris spread out from the mouths of these canyons, forming the continental rise that links the base of the continental slope with the broad, flat abyssal plains of the deep ocean floor.

ICE RAFTS ►
In the polar regions, glaciers and huge continental ice sheets flow right down to the coast. They carry heavy loads of ground-up rock and shattered rock fragments that have been plucked from the underlying rock by the moving ice. The ice can be almost black with the minerals that it contains. When the glaciers reach the sea they break up into icebergs. These float away and gradually melt, dumping their rocky and muddy loads into the water, where it settles on the ocean floor. This process has helped to form the broad abyssal plains around Antarctica.